

(g) Quality assurance		
2g.01	Know that a set of sample means are more closely distributed than individual values from the same population.	e.g. the set of mean heights from each class in a school will show less variation than the set of heights of all students in the school.
2g.02	Use action and warning lines in quality assurance sampling applications.	Control charts used for sample mean, median or range is expected. For example, in a manufacturing process to test that certain measurements are within allowable limits. Understand that almost all means, medians or ranges fall inside the action lines (action limits), and only 1 in 20 fall outside the warning lines (warning limits). Know that warning lines are set at ± 2 standard deviations of the sample mean from the expected value, and action lines are set at ± 3 standard deviations of the sample mean from the expected value. Know the action to be taken if a sample value falls outside each type of limit.